



DATA STRUCTURES AND ALGORITHMS

CERTIFICATION COURSE

**Master Data Structures and Algorithms
to Take Your Career to the Next Level!**

Data Structures and Algorithm

The Data Structures and Algorithms course, offered by W3 Technologies, will help you confidently solve complex computational problems using key data structures and algorithms. You'll learn to efficiently organize and process data with structures like arrays, linked lists, trees, and graphs, while focusing on algorithmic techniques like sorting, searching, and recursion. With a strong emphasis on optimization, you'll develop efficient, scalable solutions for real-world applications.

Why W3 Technologies

Training at W3 Technologies equips you to excel in today's competitive world. From the very beginning of your journey with us, you'll experience continuous growth through learning. We provide support throughout the course, while encouraging you to become independent and confident in your skills.



Lower Learning Cost

Learning with us is budget-friendly. Our courses are affordable yet provide comprehensive and engaging content.



Expert Instructors

Learning with us means benefiting from top-tier expertise. Our expert instructors provide deep knowledge and engaging lessons.



Certification

Earn an industry-recognized certification upon completing the course, enhancing your resume.

Introduction

- What are Data Structures?
- What is an algorithm?
- Why are Data Structures and Algorithms important?
- Types of Data Structures Types of Algorithms

Big O Notation

- Big O, Big Theta and Big Omega
- Time complexity examples
- Space Complexity Data types
- How to measure the codes using Big O?
- Relationships Between Various Notations
- Best, Worst and Average Case Analysis

Arrays

- Array
- One-Dimensional Array
- Multi-Dimensional Array
- Dynamic Arrays
- Why do Arrays fail?
- Problems Solving in Array

Linked List

- Singly Linked List
- List(Insertion, Deletion and Traversal)
- Doubly Linked List
- Circular Linked List
- Arrays Vs Linked List

Stack

- Introduction to Stack Properties
- Push and Pop
- Expression Evaluation (Infix, Prefix and Postfix)

Queue

- Why do we need Queue?
- Enqueue and Dequeue
- Linear and Circular Queue
- Practise Question on Queues and Stack

Searching Algorithms

- Introduction to Searching Algorithms Linear Search
- Time Complexity of Linear Search
- Binary Search
- Time Complexity of Binary Search

Sort Algorithms

- What is Sorting?
- Types of Sorting
- Sorting Terminologies
- Bubble Sort
- Time Complexity of Bubble Sort
- Selection Sort
- Time Complexity of Selection Sort
- Insertion Sort
- Time Complexity of Insertion Sort
- Merge Sort
- Time Complexity of Merge Sort
- Quick Sort
- Time Complexity of Quick Sort
- Heap Sort
- Time Complexity of Heap Sort
- Comparison of Sorting Algorithms

Trees Logical

- Structure and Implementations
- Terminologies and Traversals
- Types of Binary Search Tree
- Properties of a Tree: Depth, Nodes and Leafs Maximum
Depth of a Binary tree

Balanced Trees: AVL Trees

- AVL Trees: What and Why?
- Hashing
- Hash Tables
- Direct access table
- Hash Function and collisions
- Chaining and Load Factor

Graphs Algorithm

- Graph Representation Graph Traversal
- Adjacency Matrix
- Graph traversal – BFS
- Graph Traversal – DFS
- BFS Traversal vs DFS Traversal
- Minimum Spanning Tree
- Kruskal's Algorithm
- Prim's Algorithm

Shortest Path

- Dijkstra's Algorithm Importing
- Bellman Ford Algorithm

Recursion

- Recursion Programming
- Recursion Tree Method
- Master Theorem
- Extended Master Theorem
- Cracking Recursion Interview Question

Dynamic Programming

- What is Dynamic Programming ?
- Where does the name of DC come from ?
- Top Down with Memoization
- Bottom Up with Tabulation
- Top Down vs Bottom Up
- Is Merge Sort Dynamic Programming ?
- Number Factor Problem using Dynamic Programming
- Longest Common Subsequence
- Knapsack
- Bellman Ford Algorithm as Dynamic Programming
- Matrix Chain Multiplication
- Subset Sum Problem
- Floyd Marshall Algorithm

Greedy Algorithm

- Introduction
- Activity Selection Problem
- Fractional Knapsack Problem
- Huffman Coding For Data Compression
- Minimum Spanning Tree: Greedy Kruskal's Algorithm
- Greedy Algorithm: Dijkstra's Algorithm

Thank You for Choosing **W3 Technologies**

Thank you for choosing to explore this course from W3 Technologies. We hope you find it insightful and valuable as you embark on your learning journey.

At W3 Technologies, we believe that learning is a continuous process, and we are excited to be part of your journey towards gaining new skills and knowledge.



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